



Graphics Cards

Connectors



DVI and D-Sub 15pin

D-Sub (top) was the traditional connect for PC monitors and is an analog connect. DVI (bottom) is the new digital connect for monitors and is mainly found on LCD monitors

HDMI

HDMI (High-Definition Multimedia Interface) is a new interface mainly meant for home entertainment systems. It is a digital interface that is identical to DVI in every way except that it carries both audio and video signals. HDMI is seeing use in large screen plasma and LCD displays and some graphics cards and most integrated graphics solutions these days sport them. Although this standard hasn't seen large scale adoption due to low acceptance and the issues that plague DRM (Digital Rights Management) HDMI should become the standard for high-definition interconnectivity in the future.



Display-Port

DisplayPort is another newer digital interface standard that was approved in May 2006. Its main advantages are its license and royalty free. DisplayPort has hardly pervaded graphics cards today and we see very few cards that support this standard.



S-Video

S-Video is the other video connect that is used. This is an older standard and is analog in nature. It carries two separate video signals luma (luminance) and chroma (colour). It does not carry audio.



The graphics industry does a refresh every six months or so; which is annoying because if you want to play the latest games at highest resolutions and maximum detail levels you should ideally upgrade your graphics card every year. It seems the industry is slowing down, much to the joy of most people who invest upward of Rs 15,000 on a graphics card. However, be prepared for the unexpected flip — where hardware technology has a sudden bump in performance and your ten month old card is little more than a hunk of junk.

We divide our readers and consumers of discrete graphics technology into three main chunks, on the basis of how much they will / can spend on obtaining a graphics card. In India 95 per cent of the PC users use integrated graphics. Out of the five per cent of PC users who purchase graphics cards we estimate 95 per cent of them are willing to spend no more than Rs 5,000; which reinforces the fact that India is still a very price conscious market. The good news is that the number of people spending / willing to spend more than Rs 10,000 on a graphics card has risen sharply over the past two years. As 3D games become a pastime for more people, this number should increase further.

If you're a person making his first foray into the immersive world of discrete graphics you will get intimidated by the sheer number choices around. Terms like stream processors, video memory, GPU clocks and GDDR3 will float around you till you wish someone would build a dictionary filled with explanations of such terms.

Video Memory

Video memory also known as VRAM is one of the most essential performance affecting components inside your graphics card. Video memory is used to hold information necessary for a graphics card to drive a monitor / display device. Although non-gamers need no more than 128 MB of video RAM. For gaming you may need as much as 1 GB of video memory; especially if you are into gaming at high resolutions and at maximum settings. Low-end cards do not do much with more video memory.

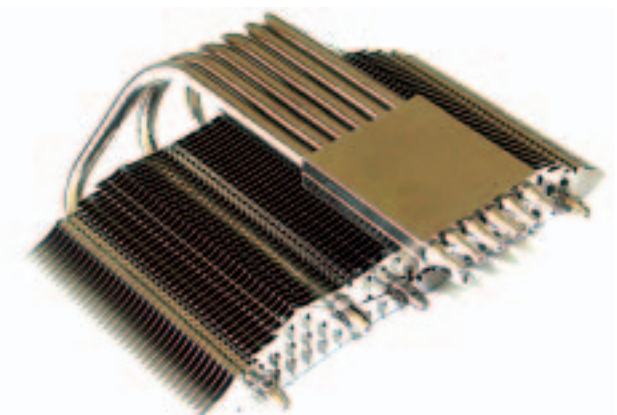


Cooling Solution

Your graphics card generates a lot of heat although the amount of heat generated depends on the GPU core and its power consumption. A high-end GPU would obviously consume more power and therefore generate more heat. There are also third party cooling solutions available that you can slap on to your graphics card if the stock cooler isn't doing its job as well. Heat pipe based solutions generally work well. Brands like Thermalright and CoolerMaster are available in India



GPU cooling



A cooler developed by Artic and meant for older ATI cards

Stream Processors

Previously known as shader units and referred to as SP(s), these refer singularly to a single processing core sitting atop a GPU. A single SP is useless on its own. Collectively these work in cohesion and are what power the display on your PC, think of them as multiple computational units working together on a variety of image and video calculations. So the more SPs (in general) the more powerful a GPU is. Although both players (ATI and NVIDIA) have different approaches to stream processing, the underlying principle remains the same; that is parallelism.